

Approximating the Score Function using Optimal Transport

Curves and Surfaces 2026

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Laboratoire de Mathématiques d'Orsay

June 11th

(Badly and Slowly) Approximating the Score Function using Optimal Transport

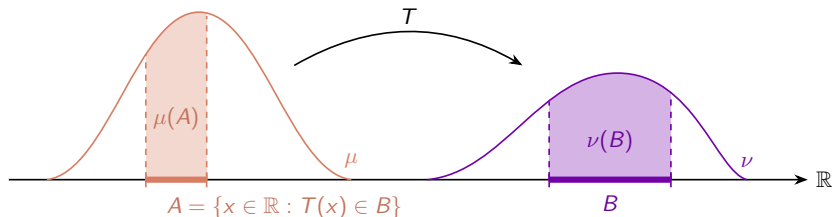
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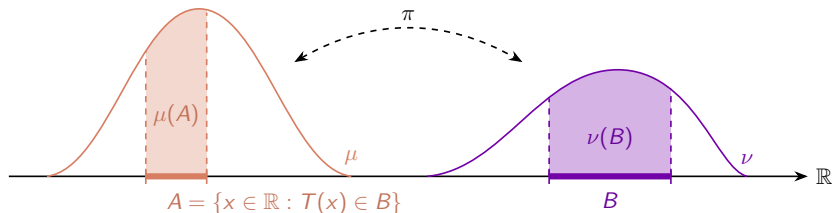
Quick reminders on Optimal Transport 1/2



- ▶ Given ground cost $c(x, y)$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
- ▶ Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x)) \mu(dx)$$

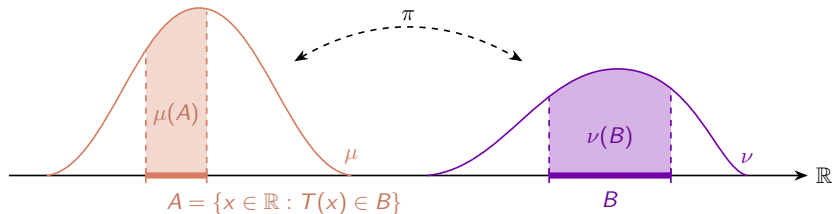
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- ▶ Given ground cost $c(x, y)$
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Among all **transport plans** π with marginals μ and ν
- ▶ Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x)) \mu(dx) \longrightarrow \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \pi(dx, dy)$$

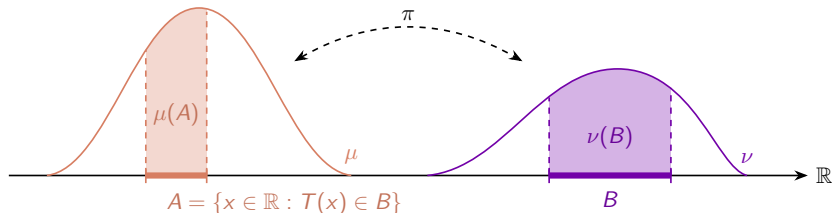
Quick reminders on Optimal Transport 2/2



► p -Wasserstein distance:

$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(dx, dy)$$

Quick reminders on Optimal Transport 2/2



- p -Wasserstein distance:

$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(dx, dy)$$

- Kantorovich duality

$$\inf_{\pi \in \Pi(\mu, \nu)} \int c(x, y) \gamma(dx, dy) = \sup_{\phi} \int \phi(x) \mu(dx) + \int \phi^c(y) \nu(dy),$$

where $\phi^c(y) := \inf_x c(x, y) - \phi(x)$.

What is the score function?

- ▶ **Forward Process (Heat Dissipation):** Running the SDE forward diffuses our data into pure noise. The density follows a Fokker-Planck equation (akin to the heat equation):

$$dx_t = f(x_t, t) dt + g(t) dw_t$$

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- ▶ **Reverse Process (Reversing Entropy):** To reverse this natural dissipation, we must inject a highly specific, restoring drift back into the system:

$$dx_t = \left[f(x_t, t) - g(t)^2 \underbrace{\nabla \log p_t(x_t)}_{\text{The Score Function}} \right] dt + g(t) d\bar{w}_t$$

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 - ▶ We can view the system through a potential energy landscape: $E_t(x) = -\log p_t(x)$.
 - ▶ The score is simply the negative gradient of this energy: $\nabla \log p_t(x) = -\nabla E_t(x)$.

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 - ▶ We can view the system through a potential energy landscape: $E_t(x) = -\log p_t(x)$.
 - ▶ The score is simply the negative gradient of this energy: $\nabla \log p_t(x) = -\nabla E_t(x)$.
 - ▶ **Conclusion:** The score acts as the exact thermodynamic **force** required to fight the heat equation and push the dispersed particles back into their highly-structured original states.

Using OT to learn the score

Consider $v_t = -\nabla(\ln \rho_t + V)$ where $V(x) = \frac{\|x\|^2}{2}$. Then, the Fokker-Planck equation

$$\partial_t \rho_t = \Delta \rho_t + \operatorname{div}(\rho_t \nabla V).$$

¹R. Jordan, D. Kinderlehrer, and F. Otto. "The Variational Formulation of the Fokker-Planck Equation". en. In: *SIAM Journal on Mathematical Analysis* 29.1 (Jan. 1998), pp. 1–17. ISSN: 0036-1410, 1095-7154. DOI: 10.1137/S0036141096303359. URL: <http://epubs.siam.org/doi/10.1137/S0036141096303359>.

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- ▶ transports any initial condition $\rho_0 \in \mathcal{P}(\mathbb{R})$ towards the standard Gaussian $\gamma = \mathcal{N}(0, Id)$ and
- ▶ can be seen as the Wasserstein gradient flow of the energy¹

$$\mathcal{F}(\rho) := \int V \rho + \int \rho \ln \rho$$

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Using OT to learn the score

We can discretize in time the gradient flow using implicit Euler scheme (or **JKO scheme**):

$$\rho_{n+1}^\tau := \arg \min_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2(\rho, \rho_n^\tau)^2 \right\} \quad (1)$$

²F. Santambrogio. *Optimal Transport for Applied Mathematicians*. Vol. 87. Progress in Nonlinear Differential Equations and Their Applications. Springer, Oct. 2015. ISBN: 978-3-319-20827-5.

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As explained by Santambrogio in his book², we have

$$\nabla \psi_n^{c_\tau} = v_{n\tau} = -\nabla(\ln \rho_{n+1}^\tau + V),$$

where $\psi_n^{c_\tau}$ is a Kantorovich potential related to the transport from ρ_n^τ to ρ_{n+1}^τ . Therefore approximating the score amounts to approximating $\nabla \psi_n^{c_\tau}$.

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Stochastic Gradient Descent

Theorem 1

If $\alpha_k = \frac{R}{B\sqrt{k}}$, then

$$\forall T \geq 1, \quad \mathbb{E} \left[G_n^T(\bar{\theta}_T^n) - G_n^T(\theta_n^T) \right] \leq BR \left(\frac{2}{T} + \frac{1}{\sqrt{T}} \right),$$

where $\bar{\theta}_T^n$ denotes the arithmetic mean of the family $(\theta^{(k)})_{k=1}^T$ and B depends only on R and $\sup_x \|\phi(x)\|$.

Some animations

[Click here to play video](#)

Thank you for your attention!